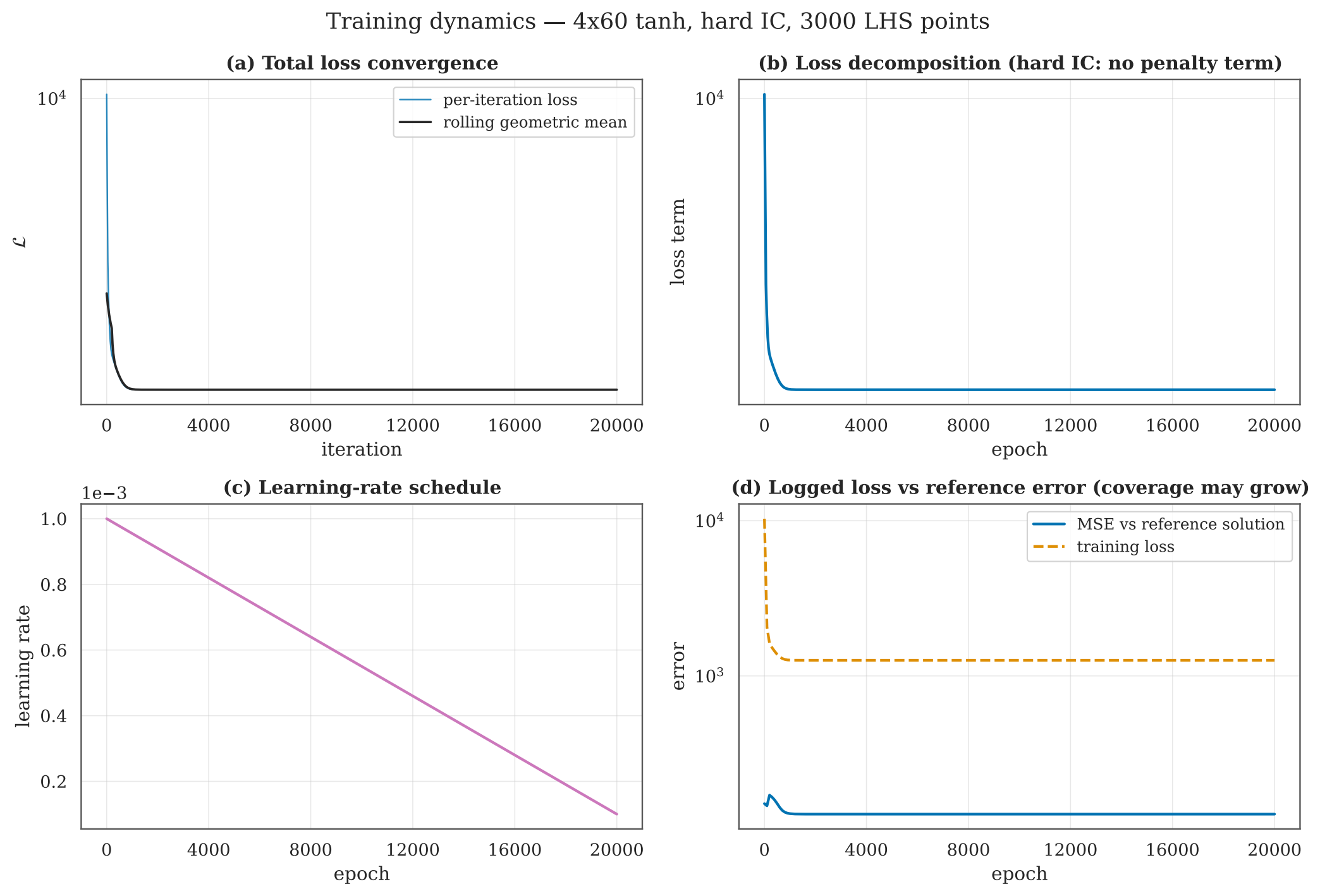
Training dynamics — 4x60 tanh, hard IC, 3000 LHS points
(a) Total loss convergence
per-iteration loss
rolling geometric mean
$10^4$
$\mathcal{L}$
0
4000
8000
12000
16000
20000
iteration
(b) Loss decomposition (hard IC: no penalty term)
$10^4$
loss term
0
4000
8000
12000
16000
20000
epoch
(c) Learning-rate schedule
1e−3
1.0
0.8
0.6
0.4
0.2
learning rate
0
4000
8000
12000
16000
20000
epoch
(d) Logged loss vs reference error (coverage may grow)
MSE vs reference solution
training loss
$10^4$
$10^3$
error
0
4000
8000
12000
16000
20000
epoch